

Diet for High Cholesterol: Recommended Foods, Patterns, and Foods to Avoid

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Background: diet as first-line management for high cholesterol

Diet and other lifestyle changes are a standard first step for managing high cholesterol. The focus has shifted from just "low fat" to an individualized, whole-food eating pattern that helps improve blood lipids, weight, and overall heart risk. (3 sources)

Diet is widely recommended as part of first-line treatment for high cholesterol, along with weight management and physical activity. Earlier cholesterol guidance emphasized a low-fat, low-cholesterol diet as part of therapeutic lifestyle changes, while newer nutrition guidance puts more weight on individualized eating patterns built around nutrient-rich, minimally processed whole foods in sensible portions. This shift matters because cholesterol management is not only about one nutrient; it is about a sustainable way of eating that improves blood lipids and overall cardiovascular risk over time. In practice, commonly recommended patterns include the Mediterranean diet and the DASH diet, both of which fit this whole-food approach and can support healthier cholesterol levels. ([Hayes et al., 2016](#)) ([Talwalkar et al., 2013](#)) ([Evert et al., 2013](#))

Evidence

(Hayes et al., 2016)

[What the Latest Evidence Tells Us About Fat and Cardiovascular Health](#)

"The first report of the National Cholesterol Education Program, released in 1988, encouraged therapeutic lifestyle changes (nutrition, weight management, and physical activity) as first-line therapy for treating high blood cholesterol. Included in these recommendations was advice to eat a low-fat, low-cholesterol diet (Talwalkar et al., 2013)"

".Current diabetes nutrition recommendations center around individualized eating patterns that focus on a nutrient-rich, whole-foods approach (in appropriate portions) to attain individual blood glucose, blood pressure, blood lipid, and body weight goals, as well as to delay or prevent complications (6,(Evert et al., 2013)"

".Two evidence-based eating patterns often recommended for managing diabetes and lowering CVD risk include the Mediterranean and Dietary Approaches to Stop Hypertension (DASH) diets."

(Talwalkar et al., 2013)

[Journey in guidelines for lipid management: From adult treatment panel \(ATP\)-I to ATP-III and what to expect in ATP-IV](#)

"Adult Treatment Panel (ATP), an expert panel to supervise cholesterol management was set up under the aegis of National Cholesterol Education Program (NCEP) in 1985. Since then NCEP-ATP has been revising and framing guidelines to enable clinician to deliver better treatment to cardiovascular patients and to educate general people. As a result, considerable reduction in cardiovascular related deaths has been observed in recent times. All three ATP guidelines viz. ATP-I, ATP-II and ATP-III have targeted

low density lipoprotein as their primary goal. The ATP-III guideline was updated in the light of evidences from 5-major clinical trials and was released in 2004. It added therapeutic lifestyle changes, concept of risk equivalents, Framingham CHD-risk score non-high density lipoprotein cholesterol (non-HDL-C) as secondary target and gave strong emphasis on metabolic risk factors. The earlier treat-to-target paradigm faced fierce criticism from clinicians across the globe because of insufficient proof of safety and benefits of treating patients with respect to an individual's low density lipoprotein (LDL) level. Further, demonstration of non-HDL-C and total cholesterol/HDL-C ratio as strong predictors of overall cardiovascular risk foresees new guidelines. A tailored-treatment approach was suggested instead of LDL-C target based treatment approach which was soundly based on direct clinical trials evidences and proposes treatment based on individual's overall 5- to 10-year cardiovascular risk irrespective of LDL-C level, leading to lower number of people on high dose/s of statins. Recent report of the Cholesterol Treatment Trialists' Collaborators meta-analysis strongly supported primary prevention of LDL with statins in low risk individuals and showed that its benefits completely outweighed its known hazards. Markers other than LDL-C like apolipoprotein B, non-HDL-C and total cholesterol/HDL-C ratio would take precedence in the risk assessment and strong emphasis would be given on tailored-treatment approach in the upcoming ATP-IV guideline."

(Evert et al., 2013)

Nutrition Therapy Recommendations for the Management of Adults With Diabetes

"There is no standard meal plan or eating pattern that works universally for all people with diabetes. In order to be effective, nutrition therapy should be individualized for each patient/client based on his or her individual health goals; personal and cultural preferences; health literacy and numeracy; access to healthful choices; and readiness, willingness, and ability to change. Nutrition interventions should emphasize a variety of minimally processed nutrient dense foods in appropriate portion sizes as part of a healthful eating pattern and provide the individual with diabetes with practical tools for day-to-day food plan and behavior change that can be maintained over the long term."

Recommended dietary patterns and nutrient targets for high cholesterol

The best-supported eating patterns for high cholesterol are Mediterranean-style, DASH-style, vegetarian, and Portfolio-style diets built around whole plant foods, fish, and healthy fats. Most guidance agrees on cutting saturated fat, avoiding trans fat, keeping dietary cholesterol low, and adding soluble fiber and plant sterols. (24 sources)

Rather than focusing on a single "cholesterol diet," most guidance supports a few similar eating patterns that lower LDL and improve overall heart risk: Mediterranean-style diets, DASH-style diets, vegetarian patterns, and the more targeted Portfolio diet. Mediterranean and vegetarian patterns can improve lipid profiles, and high-glycemic diets tend to worsen them. (Pawlik-Sobecka et al., 2019) A Mediterranean diet is widely recommended for people with dyslipidemia and ASCVD prevention, with high use of vegetables, fruits, whole grains, legumes, and olive oil, moderate fish or seafood, and lower intake of red and processed meats. (Formisano et al., 2021) (Lacatusu et al., 2019) DASH-style eating can also fit LDL-lowering advice when it emphasizes vegetables, fruits, whole grains, low-fat dairy, legumes, poultry, fish, nuts, and limited sweets and red meat. (Le et al., 2016) (Kohli et al., 2014) The Portfolio diet goes a step further by combining cholesterol-lowering foods such as nuts, plant proteins, viscous fiber, and plant sterols on top of a low-saturated-fat base, and it has shown clinically meaningful LDL reduction in hyperlipidemia. (Ferro et al., 2020) (Chiavaroli et al., 2018)

Across these patterns, the nutrient targets are fairly consistent. A common goal is to keep saturated fat below 10% of calories, and below 7% if LDL is high or hypercholesterolemia is present; trans fat should be avoided. (Hussein et al., 2013) (Lim et al., 2019) (Kim et al., 2021) (Krehic, 2024) Several guidelines also limit dietary cholesterol to less than 300 mg/day, with a stricter target of less than 200 mg/day often used when cholesterol remains elevated. (Bekele, 2020) (Kang et al., 2021) (DiLauro et al., 2023) (Kim, 2025) Replacing saturated fat with unsaturated fats, especially monounsaturated fats and omega-6 polyunsaturated fats, can help lower LDL. (Soran et al., 2017) (Kim et al., 2021) This is why Mediterranean-style fat intake can still work well even when total fat is not very low, as long as most of that fat comes from olive oil, nuts, seeds, and fish rather than butter or fatty meats. (Jimenez-Torres et al., 2021) (Ferro et al., 2020)

Fiber targets also matter. Many recommendations call for at least 25 g/day of total fiber, with 10–25 g/day coming from viscous or soluble fiber such as oats, barley, legumes, and some fruits, because this can reduce LDL. (Lim et al., 2019) (Ito et al., 2009) (Kim et al., 2021) Plant sterols or stanols at about 2 g/day are another common add-on target for lowering LDL. (Lim et al., 2019) (Elisa et al., 2022) Mediterranean guidance also often aims for roughly 45–55% of calories from carbohydrates, favoring whole grains and other high-fiber

sources rather than refined starches, with total dietary fiber around 25–40 g/day. ([Formisano et al., 2021](#))

For people with mixed dyslipidemia or high triglycerides along with high cholesterol, diet quality matters as much as fat quantity. Guidance commonly advises limiting simple sugars and refined carbohydrates and avoiding very low-fat, high-carbohydrate diets in people with insulin resistance or metabolic syndrome, because replacing saturated fat with unsaturated fat works better than replacing it with refined carbs. ([Shao et al., 2011](#)) ([Le et al., 2016](#)) ([Sawhney et al., 2023](#)) In practical terms, an effective target pattern is a mostly whole-food diet centered on vegetables, fruits, legumes, whole grains, nuts, seeds, fish, and lean or plant proteins, with low saturated fat, no trans fat, limited dietary cholesterol, and regular use of soluble-fiber foods and, if needed, plant sterol products. ([Petersen et al., 2021](#)) ([Datta, 2013](#)) ([Baratta et al., 2023](#))

Evidence

([Pawlik-Sobecka et al., 2019](#))

[Exogenous dietary factors as important modulator of human lipid profile](#)

"The most positive effect demonstrates an appropriate balanced diet, such the vegetarian and Mediterranean diets, which can significantly improve lipid profile parameters"

".The most negative effect on the all lipid profile parameters is connected with nutritional factors derived from the high glycaemic index diet."

([Formisano et al., 2021](#))

[Effects of a Mediterranean Diet, Dairy, and Meat Products on Different Phenotypes of Dyslipidemia: A Preliminary Retrospective Analysis](#)

"A Mediterranean diet (MD) is the main dietary model recommended for the prevention of ASCVD ([Mattioli et al., 2017](#)) and is the reference diet model of the 2019 European Society of Cardiology/European Atherosclerosis Society (ESC/EAS) guidelines for the management of dyslipidemia (2020). In patients affected by hyperlipidemia, the MD recommends low intake of saturated fatty acids (SFAs), at least less than 10% of total energy intake (i.e., <7% in patients with hypercholesterolemia). Consequently, moderate restriction of milk and dairy product consumption should be balanced by a limited intake of meat and meat products ([Lacatusu et al., 2019](#)), in particular, the preferred consumption of milk and dairy products should not exceed 180 g/day, while no more than 80 g/day of meat and meat products should be consumed ([Sofi et al., 2017](#)). Low-fat cheeses and semi-skimmed milk should be preferred for patients with dyslipidemia ([Drouin-Chartier et al., 2016](#)), and processed meats should not be recommended ([Simpson et al., 2019](#)). A high prevalence of plant-based food, such as whole grains, vegetables, and fruits are highly advisable according to the MD in order to reach a total amount of carbohydrates between 45 and 55% of total energy intake, and 25-40 g per day of total dietary fiber [8]. Furthermore, the MD encourages a moderate amount of seafood, regular consumption of olive oil, and increased physical activity [13]14]."

([Lacatusu et al., 2019](#))

[The Mediterranean Diet: From an Environment-Driven Food Culture to an Emerging Medical Prescription](#)

"The Mediterranean diet originates in the food cultures of ancient civilizations which developed around the Mediterranean Basin and is based on the regular consumption of olive oil (as the main source of added fat), plant foods (cereals, fruits, vegetables, legumes, tree nuts, and seeds), the moderate consumption of fish, seafood, and dairy, and low-to-moderate alcohol (mostly red wine) intake, balanced by a comparatively limited use of red meat and other meat products. A few decades ago, the Mediterranean diet drew the attention of medical professionals by proving extended health benefits. The first reports ascertained cardiovascular protection, as multiple large-scale clinical studies, starting with Ancel Keys' Seven Countries Study, showed a marked reduction of atherosclerotic clinical events in populations with a Mediterranean dietary pattern. Ensuing trials confirmed favorable influences on the risk for metabolic syndrome, obesity, type 2 diabetes mellitus, cancer, and neurodegenerative diseases. While its health benefits are universally recognized today by medical professionals, the present state of the Mediterranean diet is challenged by major difficulties in implementing this protective dietary pattern in other geographical and cultural areas and keeping it alive in traditional Mediterranean territories, also tainted by the unhealthy eating habits brought by worldwide acculturation."

([Le et al., 2016](#))

[Effects of Diet Composition and Insulin Resistance Status on Plasma Lipid Levels in a Weight Loss Intervention in Women](#)

"For patients with metabolic syndrome, a condition of insulin resistance, replacing saturated fat with unsaturated fat rather than carbohydrate and avoiding very low-fat diets has been recommended to avoid further worsening high triglyceride and low high-density lipoprotein cholesterol (HDL-C) levels"

"urrent dietary guidelines for lowering LDL-C recommend limiting the intake of saturated fat by adhering to either a higher fat (eg, Mediterranean) or lower fat (eg, Dietary Approaches to Stop Hypertension, or DASH) dietary pattern. 1"

(Kohli et al., 2014)

Clinician's Guide to the Updated ABCs of Cardiovascular Disease Prevention

"The guideline recommended that patients eat plenty of vegetables, fruits, and whole grains; incorporate low-fat dairy products, poultry, fish, legumes, nontropical vegetable oils, and nuts into their diet; and limit intake of sweets, sugar-sweetened beverages, and red meats. Patients that specifically need to lower their cholesterol levels should reduce saturated and trans fat consumption, with ideally only 5% to 6% of daily calorie intake coming from saturated fat"

". The Mediterranean-style diet (enriched with olive oil, legumes, fish, chicken, nuts, wine, fruits and vegetables and low in artificial sugars, commercial sweets, pastries, butter, margarine, and red meat) also yields heart-healthy benefits. 137"

(Ferro et al., 2020)

Effects of a Portfolio-Mediterranean Diet and a Mediterranean Diet with or without a Sterol-Enriched Yogurt in Individuals with Hypercholesterolemia

"The Mediterranean Diet (MedDiet) is associated with a reduced low density lipoprotein cholesterol (LDL-C), even in the absence of weight loss (Richard et al., 2011)(Escuriol et al., 2009)"

".LDL-C can be reduced further through a combination of ingredients with cholesterol-lowering properties, such as plant sterols/ stanols, viscous plant fibres from oats, barley, and nuts, soymilk, and soy meat analogues. This leads to an LDL-C reduction from 20% to 30% and a triglyceride reduction of 16% in individuals with hyperlipidemia compared with diet alone (Chiavaroli et al., 2018)."

(Chiavaroli et al., 2018)

Portfolio Dietary Pattern and Cardiovascular Disease: A Systematic Review and Meta-analysis of Controlled Trials.

"BACKGROUND

The evidence for the Portfolio dietary pattern, a plant-based dietary pattern that combines recognized cholesterol-lowering foods (nuts, plant protein, viscous fibre, plant sterols), has not been summarized.

OBJECTIVE

To update the European Association for the Study of Diabetes clinical practice guidelines for nutrition therapy, we conducted a systematic review and meta-analysis of controlled trials using GRADE of the effect of the Portfolio dietary pattern on the primary therapeutic lipid target for cardiovascular disease prevention, low-density lipoprotein cholesterol (LDL-C), and other established cardiometabolic risk factors.

METHODS

We searched MEDLINE, EMBASE, and The Cochrane Library through April 19, 2018. We included controlled trials ≥ 8 weeks assessing the effect of the Portfolio dietary pattern on cardiometabolic risk factors compared with an energy-matched control diet free of Portfolio dietary pattern components. Two independent reviewers extracted data and assessed risk of bias. The primary outcome was LDL-C. Data were pooled using the generic inverse-variance method and expressed as mean differences (MDs) with 95% confidence intervals (CIs). Heterogeneity was assessed (Cochran Q statistic) and quantified (I²-statistic). GRADE assessed the certainty of the evidence.

RESULTS

Eligibility criteria were met by 7 trial comparisons in 439 participants with hyperlipidemia, in which the Portfolio dietary pattern was given on a background of a National Cholesterol Education Program (NCEP) Step II diet. The combination of a portfolio dietary pattern and NCEP Step II diet significantly reduced the primary outcome LDL-C by $\sim 17\%$ (MD, -0.73 mmol/L, [95% CI, -0.89 to -0.56 mmol/L]) as well as non-high-density lipoprotein cholesterol, apolipoprotein B, total cholesterol, triglycerides, systolic and diastolic blood pressure, C-reactive protein, and estimated 10-year coronary heart disease (CHD) risk, compared with an NCEP Step 2 diet alone ($p < 0.05$). There was no effect on high-density lipoprotein cholesterol or body weight. The certainty of the evidence was high for LDL-cholesterol and most lipid outcomes and moderate for all others outcomes.

CONCLUSIONS

Current evidence demonstrates that the Portfolio dietary pattern leads to clinically meaningful improvements in LDL-C as well as other established cardiometabolic risk factors and estimated 10-year CHD risk."

(Hussein et al., 2013)

Transcultural Diabetes Nutrition Algorithm: A Malaysian Application

"For T2D patients with dyslipidemia, lifestyle modification focusing on the reduction of saturated fat (<7% of total calories), trans fat (avoid), and cholesterol (<200 mg/day) intake has been recommended [12]19]."

(Lim et al., 2019)

Nutritional Therapy for Asian Patients at Risk for Atherosclerotic Cardiovascular Disease

"The therapeutic lifestyle changes (TLC) diet was developed by the Adult Treatment Panel III of the National Cholesterol Education Program Guidelines (2013). (Eckel et al., 2014) In the TLC diet, saturated fat accounts for <7% of the total calories recommended, and cholesterol intake is <200 mg/day. Increases in viscous (soluble) fiber (10-25 g/day) and plant stanol/sterol (2 g/day) intake are recommended to reduce LDL-C, as are weight management and increased physical activity"

".The American College of Cardiology/American Heart Association (ACC/AHA) task force (Horn et al., 2016) recommends consumption of cardioprotective foods such as plant proteins (e.g., soy protein, nuts, and legumes), plant sterols, fiber, and soluble fiber-containing nutrients to promote heart health. MED and vegetarian (portfolio) diets that encourage consumption of vegetables, fruits, whole grains, and fish can improve heart health and lower cholesterol levels."

(Kim et al., 2021)

Lifestyle Modification in the Management of Metabolic Syndrome: Statement From Korean Society of CardioMetabolic Syndrome (KSCMS)

"Thus, it is recommended to keep saturated fatty acids <10% of total energy and should be further reduced (<7% of energy) in the presence of hypercholesterolemia. (2020) The evidence relevant for the intake of unsaturated fatty acids in dyslipidemia is not yet sufficient, but LDL cholesterol levels decrease when saturated fatty acids are replaced with monounsaturated fatty acids or omega-6 polyunsaturated fatty acids. (Rodríguez, 2014) Trans-fat intake is associated with increased LDL:HDL cholesterol ratio and cardiovascular risk. (Mozaffarian et al., 2006) Hence, it is advisable to avoid transfat intake. A daily fiber intake of 25 g or more is recommended since dietary intake of fibers can reduce the risk of cardiovascular disease by decreasing blood glucose and LDL cholesterol levels. 72)104)"

(Krehic, 2024)

Influence of Modified Diet and Low Grade Physical Activity on Serum Lipids in Patients With Type IIa and IIb dyslipoproteinaemia

"Diet is the treatment of first choice in non-pharmacological treatment of elevated serum cholesterol values"

"It is based on reducing the intake of saturated fatty acids from food of animal origin, increasing the intake of fruits, vegetables and fish (Dwyer, 1997;Diab, Dastmalchi, Gulati and Michos, 2023)"

".Step I diet implies intake of saturated fatty acids in an amount that corresponds to 8 to 10% of total calories, 30% and fewer calories from total fats and cholesterol less than 300 mg/day. If the desired results are not achieved with the diet in Step I, it is switched to Step II, which implies intake of saturated fatty acids in an amount less than 7% of total calories and cholesterol less than 200 mg/ day."

(Bekele, 2020)

A Review on the Relationship Between Nutrition and Health Condition in Humans

"The National Cholesterol Education Program (Clark, 1992) encourages people with high total cholesterol and low-density lipoprotein (LDL) cholesterol to start with a step I diet that has less than 30% of the total calories coming from fat, 8% to 10% of calories from saturated fat, and less than 300 mg of cholesterol per day. If lipid levels do not decrease after three months, a step II diet is tried in which less than 30% of calories come from total fat, less than 7% from saturated fat, and less than 200 mg from cholesterol (NCEP, 1993)"

".Suggestions for lowering fat and saturated fat include: avoiding fats as spreads or for flavoring, avoiding or reducing consumption of meat, using specially manufactured low-fat foods (e.g., fat-free salad dressings), modifying common foods to be lower in fat (e.g., remove skin from chicken), and replacing a high-fat food with its low-fat equivalent (e.g., skin milk instead of whole milk)"

".More aggressive diets are also an option for those with high lipid values. These diets contain 10% of total calories from fat, less than 3% from saturated fatty acids (SFAs) and less than 5 mg of cholesterol per day with a minimal amount of animal fat"

".Nuts have been shown to lower serum cholesterol levels (Kristal et al., 1992). Walnuts lowered LDL cholesterol by about 16%, and almonds or hazelnuts reduced serum cholesterol by 8% to 10%"

".Including a 1-oz portion of nuts in a low-fat diet should pose no problem while keeping calories under control."

(Kang et al., 2021)

Relationship between Dietary Habits and Control of Lipid Profiles in Patients with Dyslipidemia Using Pravastatin

"Therefore, several guidelines have recommended limiting daily cholesterol intake to <300 mg for people with high serum cholesterol levels (Reiner et al., 2011)[6]"

".Patients with dyslipidemia should consume less cholesterol, consume more dairy and protein-rich foods, and follow a regular eating schedule to control lipid profiles."

(DiLauro et al., 2023)

The Healthy Eating Assessment Tool (HEAT): A Simplified 10-Point Assessment of CHLD-2 Dietary Compliance for Children and Adolescents with Dyslipidemia

"A low-fat, low-sugar, and high-fiber diet that relies primarily on vegetables and fruits, whole grains, lean meats, fish, lentils and legumes, and low-and non-fat dairy products is recommended"

".saturated fat is to be limited to 8-10% of daily kcal/EER, while mono and polyunsaturated fat consumption is encouraged up to 20% of daily kcal/EER. Trans fat is to be avoided as much as possible, and dietary cholesterol is to be limited to <300 mg/day. The remaining 70% of dietary energy should be composed of 15-20% proteins and 50-55% carbohydrates, with a focus on fiber-rich foods such as vegetables, fruit, and whole grains. If dyslipidemia persists after 3 months, the CHLD-2 is recommended, which suggests a further restriction of saturated fat to <7% of daily kcal/EER, monounsaturated fat to 10% of daily kcal/EER, and limitation of dietary cholesterol to <200 mg/day (Table 1)."

(Kim, 2025)

Dyslipidemia in Children and Adolescents: Current Insights and Updated Treatment Approaches

"For children and adolescents with dyslipidemia, transitioning to a low-fat diet is recommended to reduce LDL-C levels"

".CHILD 1 limits total fat intake to 25-30% of total calories, saturated fat to <10%, and cholesterol to <300 mg per day"

".CHILD 2-LDL is intended for patients with hyperlipidemia characterized by elevated LDL-C, which limits total fat intake to 25-30% of total calories, saturated fat to <7%, cholesterol to <200 mg/day, and trans-fat to <1%. CHILD 2-TG is indicated for patients with hypertriglyceridemia. This approach focuses on reducing simple carbohydrates, while increasing the amount of complex carbohydrates. CHILD 2-TG also recommends consuming dietary fish to increase omega-3 fatty acid intake."

(Soran et al., 2017)

Hypercholesterolaemia – practical information for non-specialists

"In lean patients with hyperlipidaemia, dietary saturated fat and cholesterol should particularly be avoided, because they tend to raise LDL-C"

".a number of dietary trials, particularly when monounsaturates are substituted, have shown substantially decreased CVD incidence (Estruch et al., 2013)."

(Jimenez-Torres et al., 2021)

Mediterranean Diet Reduces Atherosclerosis Progression in Coronary Heart Disease

"(1) the Mediterranean diet, with a minimum of 35% of total calories from fat (22% MUFA, 6% polyunsaturated fatty acids [PUFA], and <10% saturated fatty acids [SFA]), 15% proteins, and a maximum of 50% carbohydrates"

"In both diets, the cholesterol content was adjusted to <300 mg/day"

". patients consuming the Mediterranean diet also had a high intake of oily fish, nuts, and extra virgin olive oil, together with a low intake of harmful foods such as red/processed meats and pastries/ commercial bakery products. 16"

(Ito et al., 2009)

Maintaining Cardiovascular Health in Patients With Mixed Dyslipidemia Optimizing the Management of Hypertriglyceridemia and Non-HDL Cholesterol

"Recommended therapeutic lifestyle changes include increased physical activity (moderate exercise sufficient to expend ≥ 200 kcal per day); increased intake of omega-3 fatty acids, plant stanols/sterols, and soluble fiber; a caloric intake limited to that needed to maintain an appropriate body weight and prevent weight gain; and limited dietary intake of saturated fats, cholesterol, and trans fatty acids. Limiting saturated fat intake to < 7% of total calories and dietary cholesterol to < 200 mg per day should be the primary recommendation in all patients with elevated LDL-C levels. Other therapeutic options include intake of 2 gm per day of plant sterols/stanols and 10-25 gm per day of soluble fiber. 32(Ellis et al., 2004)"

"The American Heart Association (AHA) recommends that all individuals increase the amount of fish incorporated into the diet as a source of omega-3 fatty acids that may lower CHD risk. People without coronary disease should be consuming at least 2 servings of fish per week. In patients with preexisting CHD, the AHA recommends the equivalent of 1 gm per day of omega-3 fatty acids-a level

shown to decrease the risk of cardiovascular death."

(Elisa et al., 2022)

Efficacy of Plant Sterol-Enriched Food for Primary Prevention and Treatment of Hypercholesterolemia: A Systematic Literature Review.

"A healthy diet includes a high intake of fruit, vegetables, nuts, whole grains, lean vegetable or animal protein and fish and reduced intakes of trans fats, red meat, processed meat, refined carbohydrates and sugar-sweetened drinks"

"The National Cholesterol Education Program Adult Treatment Panel III recommends combined dietary therapy with a low rate of both saturated and total fat, a total cholesterol concentration lower than 200 mg/dL and consumption of soluble fiber and plant sterols/phytosterols (PS), a dose of 2 g/day of PSs being more effective."

(Shao et al., 2011)

Treatment of dyslipidemia in the elderly

"it is now recommended that individuals with dyslipidemia should reduce intake of saturated fat, limit the consumption of simple carbohydrates and increase the proportion of protein in their diet. [22,(Aude et al., 2004)[25]"

(Sawhney et al., 2023)

CSI clinical practice guidelines for dyslipidemia management: Executive summary

"Increased intake of combination unsaturated fats as in Mediterranean diets is recommended"

"Atherogenic dyslipidemia is best managed by restriction of excess calories, sugars, refined carbohydrates, fried foods, and trans-fats while maintaining ideal body weight. Isolated elevation of LDL cholesterol is managed by restriction of trans-fats and saturated fats"

"The healthiest oils appear to be those high in monounsaturated fats (MUFA) and PUFA with a high N-3/N-6 ratio. The combination of MUFA-and PUFA-containing oils is a good option. Trans fats should be totally avoided"

".Palm oils should be avoided"

".Refined carbohydrates, sugar-sweetened beverages, and processed foods should be strongly restricted and replaced with healthy grains and fiber."

(Petersen et al., 2021)

Diet Quality Assessment and the Relationship between Diet Quality and Cardiovascular Disease Risk

"Broadly, the guidelines consistently recommend high intakes of vegetables, fruits, wholegrains, legumes, nuts and seeds, low-fat dairy, unprocessed lean meats and poultry, and seafood, and lower intakes of sources of saturated fat, added sugar, and sodium. Differences are observed for saturated fat recommendations based on the purpose of the guidelines (i.e., general health <10% kcal, management of lipids/lipoproteins (<7%), or CVD prevention/management (5-6% kcal); however, consistently, it is recommended that saturated fat intake is limited, and sources of unsaturated fats are consumed instead [41][42](Jacobson et al., 2015)[44]."

(Datta, 2013)

PRINCIPLES AND TREATMENT APPROACHES OF DYSLIPIDAEMIAS: A REVIEW

"Current nutritional guidelines for the reduction of cardiovascular risk through lipid management recommend diets rich in fruits, vegetables, grains, legumes, high fibre cereals, low-fat dairy products, fish, lean meats, and skinless poultry"

".Whereas dietary intake of trans-fatty acids is associated with both increased LDL-C and decreased HDL-C levels"

".A patient should achieve this by limiting total fat intake to 25% to 30% of calories and saturated fat to less than 7% of calories."

(Baratta et al., 2023)

Challenges in Improving Adherence to Diet and Drug Treatment in Hypercholesterolemia Patients

"A good adherence to a Mediterranean-type low-fat diet and healthful lifestyle is associated with lower serum cholesterol levels and reduced odds of having hypercholesterolemia. Therefore, current guidelines recommend initial dietary counseling by physicians for most patients with hypercholesterolemia and referral to a registered dietitian for those with persistent hypercholesterolemia (2020)6,19]. The main dietary recommendation is to minimize the consumption of saturated fats to lower LDL-C, as well as other therapeutic dietary options, such as increasing plant stanols/sterols and increasing viscous fiber (2020)."

Foods and nutrients to emphasize

Build meals around plants: vegetables, fruits, whole grains, beans, nuts, and seeds. Use unsaturated fats like olive or canola oil, choose plant proteins often, and include fish regularly; add soluble fiber and plant sterols if you need more LDL lowering. (18 sources)

The foods most worth emphasizing are the ones that reliably improve LDL and overall heart risk when they replace foods high in saturated fat. Across guidelines, this means eating plenty of vegetables, fruits, whole grains, legumes, nuts, and seeds, with fish and other lean proteins used regularly. Low-fat dairy can fit, and many recommendations also support fermented dairy such as yogurt and cheese, while still favoring lower saturated-fat choices overall. [\(Petersen et al., 2021\)](#) [\(Nestel et al., 2022\)](#) [\(Formisano et al., 2021\)](#) [\(Lim et al., 2019\)](#)

A practical way to do this is to make whole grains and other high-fiber plant foods the base of the diet. Whole grains, multigrain foods, beans, vegetables, and fruit help raise total fiber intake, and fiber has a cholesterol-lowering effect. Several sources also specifically encourage fish, including oily or “blue-back” fish, a few times per week. [\(Rhee et al., 2019\)](#) [\(Krezel et al., 2024\)](#) [\(Ito et al., 2009\)](#) [\(Cheema, 2012\)](#)

Among fats, the key point is not to fear all fat, but to use the right kinds. Olive, canola, peanut, and other vegetable oils rich in monounsaturated or polyunsaturated fat are preferred in place of butter, cream, or animal fat. Nuts and seeds are also useful because they provide unsaturated fats and can replace less healthy snacks or meats. [\(Kasiske et al., 2004\)](#) [\(Nestel et al., 2022\)](#) [\(Turini et al., 2022\)](#) [\(Krezel et al., 2024\)](#)

Protein choices matter too. Plant proteins such as soy foods, beans, lentils, and nuts are repeatedly recommended, with fish next and poultry after that. This is one reason Mediterranean, vegetarian, and Portfolio-style eating patterns can lower cholesterol: they shift protein intake away from fatty and processed meats and toward plant foods and seafood. [\(Lim et al., 2019\)](#) [\(Nestel et al., 2022\)](#) [\(Farrer, 2018\)](#) [\(Abbasi et al., 2024\)](#)

Two food components deserve special attention because they can give extra LDL lowering on top of a generally healthy diet: soluble fiber and plant sterols or stanols. Soluble or viscous fiber from oats, barley, beans, pectin-rich foods, and similar foods is commonly recommended, and plant sterols/stanols are often suggested at about 2 g/day. Used daily and with meals, these can provide additional cholesterol lowering beyond diet alone. [\(Lim et al., 2019\)](#) [\(Ito et al., 2009\)](#) [\(Rosin et al., 2015\)](#) [\(Ferruzzi et al., 2024\)](#)

If someone wants a more targeted “foods first” approach, the best-supported combo is close to the Portfolio pattern: nuts, plant proteins such as soy, viscous fiber from oats and barley, and plant sterols/stanols. This combination has shown meaningful LDL reduction in people with hyperlipidemia, especially when layered onto an already low-saturated-fat diet. [\(Ferro et al., 2020\)](#) [\(Chiavaroli et al., 2018\)](#)

In everyday food terms, that usually means choosing oatmeal or barley, beans or lentils, vegetables and fruit at most meals, nuts or natural peanut butter in moderate portions, olive oil instead of butter, and fish more often than red meat. Soy foods can also be a useful regular addition, and some guidance suggests routine intake of soy, nuts, and moderate dairy as part of a cholesterol-conscious pattern. [\(Ferruzzi et al., 2024\)](#) [\(Chang et al., 2025\)](#) [\(Kuk, 1987\)](#)

Evidence

[\(Petersen et al., 2021\)](#)

[Diet Quality Assessment and the Relationship between Diet Quality and Cardiovascular Disease Risk](#)

"Broadly, the guidelines consistently recommend high intakes of vegetables, fruits, wholegrains, legumes, nuts and seeds, low-fat dairy, unprocessed lean meats and poultry, and seafood, and lower intakes of sources of saturated fat, added sugar, and sodium. Differences are observed for saturated fat recommendations based on the purpose of the guidelines (i.e., general health <10% kcal, management of lipids/lipoproteins (<7%), or CVD prevention/management (5-6% kcal); however, consistently, it is recommended that saturated fat intake is limited, and sources of unsaturated fats are consumed instead [41][42](Jacobson et al., 2015)[44]."

[\(Nestel et al., 2022\)](#)

Dietary patterns, dietary nutrients and cardiovascular disease.

"A diet consisting of fruits, vegetables, and whole grains is recommended. Plant-based proteins are the preferred sources of protein followed by fish and poultry. A healthy dietary pattern should minimize consumption of refined starches, added sugars (including sugar-sweetened drinks), trans fats, red meats (particularly processed meats) and sodium (salt intake). Saturated fats should be replaced with polyunsaturated and monounsaturated oils. Consumption of dairy is supported (without strong evidence to favour reduced-fat products), in particular fermented dairy such as yoghurt and cheese, but avoiding butter and cream principally in individuals with diabetes or at increased risk of CVD. Cholesterol and foods high in cholesterol, including eggs and crustaceans, should be limited for individuals with diabetes or those at increased risk of CVD."

(Formisano et al., 2021)

Effects of a Mediterranean Diet, Dairy, and Meat Products on Different Phenotypes of Dyslipidemia: A Preliminary Retrospective Analysis

"A Mediterranean diet (MD) is the main dietary model recommended for the prevention of ASCVD (Mattioli et al., 2017) and is the reference diet model of the 2019 European Society of Cardiology/European Atherosclerosis Society (ESC/EAS) guidelines for the management of dyslipidemia (2020). In patients affected by hyperlipidemia, the MD recommends low intake of saturated fatty acids (SFAs), at least less than 10% of total energy intake (i.e., <7% in patients with hypercholesterolemia). Consequently, moderate restriction of milk and dairy product consumption should be balanced by a limited intake of meat and meat products (Lacatusu et al., 2019), in particular, the preferred consumption of milk and dairy products should not exceed 180 g/day, while no more than 80 g/day of meat and meat products should be consumed (Sofi et al., 2017). Low-fat cheeses and semi-skimmed milk should be preferred for patients with dyslipidemia (Drouin-Chartier et al., 2016), and processed meats should not be recommended (Simpson et al., 2019). A high prevalence of plant-based food, such as whole grains, vegetables, and fruits are highly advisable according to the MD in order to reach a total amount of carbohydrates between 45 and 55% of total energy intake, and 25-40 g per day of total dietary fiber [8]. Furthermore, the MD encourages a moderate amount of seafood, regular consumption of olive oil, and increased physical activity [13]14."

(Lim et al., 2019)

Nutritional Therapy for Asian Patients at Risk for Atherosclerotic Cardiovascular Disease

"The therapeutic lifestyle changes (TLC) diet was developed by the Adult Treatment Panel III of the National Cholesterol Education Program Guidelines (2013). (Eckel et al., 2014) In the TLC diet, saturated fat accounts for <7% of the total calories recommended, and cholesterol intake is <200 mg/day. Increases in viscous (soluble) fiber (10-25 g/day) and plant stanol/sterol (2 g/day) intake are recommended to reduce LDL-C, as are weight management and increased physical activity"

"The American College of Cardiology/American Heart Association (ACC/AHA) task force (Horn et al., 2016) recommends consumption of cardioprotective foods such as plant proteins (e.g., soy protein, nuts, and legumes), plant sterols, fiber, and soluble fiber-containing nutrients to promote heart health. MED and vegetarian (portfolio) diets that encourage consumption of vegetables, fruits, whole grains, and fish can improve heart health and lower cholesterol levels."

(Rhee et al., 2019)

2018 Guidelines for the Management of Dyslipidemia in Korea

"For patients with hypercholesterolemia, limit daily cholesterol intake to 300 mg"

"Eat fiber-rich foods to consume >25 g of dietary fiber"

"Eat diets rich in whole grains and multigrain, beans, vegetables, and fish"

"Eat whole grain and multigrain as the staple Eat sufficient amounts of vegetables Eat fish, particularly blue-back fish 2 to 3 times a week Eat an adequate amount of fresh fruits"

"limit saturated fats intake to <7% of total energy intake and to limit trans fat intake to <1% of total energy intake by avoiding processed foods.(Reiner et al., 2013)"

"Some rich sources of saturated fats include fats of meats, skin of poultry, butter, and palm oil. The major sources of trans fats are hydrogenated oils, such as margarine and shortening, and trans fats are produced during heating oil in high temperature."

(Krezel et al., 2024)

Analysis of Compliance with Proper Nutrition Principles in Patients with a History of Myocardial Infarction

"The diet that patients should follow in order to reduce the risk of developing and worsening CHD ought to include plenty of vegetables, fruit, whole grains, and plants, which have essential vitamins and minerals and also provide a high amount of fibre. This is because fibre has a cholesterol-lowering effect. Vegetable oils also have a similar effect and are therefore often recommended for patients at risk of MI. Meat should be consumed in moderate amounts, and it is advised to replace meat with fish. The consumption

of red meat, on the other hand, is discouraged altogether (Molero-Diez et al., 2019)(Szczepańska et al., 2023)(Bortkiewicz et al., 2019)."

(Ito et al., 2009)

Maintaining Cardiovascular Health in Patients With Mixed Dyslipidemia Optimizing the Management of Hypertriglyceridemia and Non-HDL Cholesterol

"Recommended therapeutic lifestyle changes include increased physical activity (moderate exercise sufficient to expend ≥ 200 kcal per day); increased intake of omega-3 fatty acids, plant stanols/sterols, and soluble fiber; a caloric intake limited to that needed to maintain an appropriate body weight and prevent weight gain; and limited dietary intake of saturated fats, cholesterol, and trans fatty acids. Limiting saturated fat intake to $< 7\%$ of total calories and dietary cholesterol to < 200 mg per day should be the primary recommendation in all patients with elevated LDL-C levels. Other therapeutic options include intake of 2 gm per day of plant sterols/stanols and 10-25 gm per day of soluble fiber. 32,(Ellis et al., 2004)"

"The American Heart Association (AHA) recommends that all individuals increase the amount of fish incorporated into the diet as a source of omega-3 fatty acids that may lower CHD risk. People without coronary disease should be consuming at least 2 servings of fish per week. In patients with preexisting CHD, the AHA recommends the equivalent of 1 gm per day of omega-3 fatty acids-a level shown to decrease the risk of cardiovascular death."

(Cheema, 2012)

Dietary Fats and Heart Disease: From Prenatal to Postnatal Environment

"It is recommended to limit the intake of foods high in saturated fat, trans fat and cholesterol, such as whole-milk dairy products, fatty meats, partially hydrogenated vegetable oils, fried foods and egg yolks. Instead, it is recommended to choose foods low in saturated fat, trans fat and cholesterol. This can be accomplished by enriching the diet in fruits and vegetables, grain products especially from whole grains, lentils and beans and to add either fish or flax powder (ground als) to the diet."

(Kasiske et al., 2004)

Clinical practice guidelines for managing dyslipidemias in kidney transplant patients: a report from the Managing Dyslipidemias in Chronic Kidney Disease Work Group of the National Kidney Foundation Kidney Disease Outcomes Quality Initiative

"Overall, healthy food preparation should be encouraged, such as using peanut, canola, or olive oil in cooking, since these are high in monounsaturated fats"

".Dietary recommendations may include high-protein foods, with a liberal intake of foods high in saturated fat. However, in the majority of cases, acceptable protein sources low in saturated fat should be encouraged (Table 28). Low-fat dairy products, nuts, seeds, and beans may provide protein, but potassium and phosphorus contents should still be limited."

(Turini et al., 2022)

Efficacy of Plant Sterol-Enriched Food for Primary Prevention and Treatment of Hypercholesterolemia: A Systematic Literature Review

"A healthy diet includes a high intake of fruit, vegetables, nuts, whole grains, lean vegetable or animal protein and fish and reduced intakes of trans fats, red meat, processed meat, refined carbohydrates and sugar-sweetened drinks"

".The National Cholesterol Education Program Adult Treatment Panel III recommends combined dietary therapy with a low rate of both saturated and total fat, a total cholesterol concentration lower than 200 mg/dL and consumption of soluble fiber and plant sterols/phytosterols (PS), a dose of 2 g/day of PSs being more effective. PSs (including betasitosterol, camp sterol and stigmasterol) have a chemical structure and cellular function similar to human cholesterol and are found in vegetable oils, nuts and seeds (Cusack et al., 2013)(Talati et al., 2010)."

(Farrer, 2018)

Beyond Statins: Emerging Evidence for HDL-Increasing Therapies and Diet in Treating Cardiovascular Disease

"The American Heart Association have released specific diet guidelines to prevent cardiovascular disease. Their major guidelines are to consume a variety of fruits and vegetables and grain products, including whole grains, as well as including fat-free and low-fat dairy products, fish, legumes, poultry, and lean meats [21]. They also say to maintain a desirable blood cholesterol and lipoprotein profile by limiting the intake of foods with a high content of saturated fatty acids and cholesterol and to substitute grains and unsaturated fatty acids from vegetables, fish, legumes, and nuts [21]"

".These patients transitioned from a usual diet to plant-based nutrition. They were only considered active participants if they completely refrained from dairy, fish, and meat. Whole grains, legumes, lentils, other vegetables, and fruit comprised the major portion of the diet."

(Abbasi et al., 2024)

New Insights Into the Treatment of Hyperlipidemia: Pharmacological Updates and Emerging Treatments

"Common dietary interventions for hyperlipidemia include decreasing overall food consumption and reducing portion, reducing saturated fat and cholesterol with monounsaturated and polysaturated fats by limiting intake of red meat, choosing lean meat like fish and poultry, and avoiding full-fat dairy products and processed food. Incorporating certain foods with favorable effects on the lipid profile, such as plant sterols and soluble fiber, can lead to a reduction of LDL-C levels by 6%-14%. Additionally, it is advisable to eliminate alcohol in patients with hypertriglyceridemia (Brown et al., 1999)[18]."

(Rosin et al., 2015)

Optimal Use of Plant Stanol Ester in the Management of Hypercholesterolemia

"Plant stanol ester is a well-established and widely recommended dietary measure to reduce serum cholesterol"

". For optimal cholesterol lowering aiming at a lower risk of cardiovascular disease, plant stanol ester should be used daily, in sufficient amounts, with a meal and in combination with other recommended dietary changes."

(Ferruzzi et al., 2024)

Case Report: Effects of Diet on Hypercholesterolemia

"Foods that contain phytosterols, beta-glucans, pectin, and soluble fibers should be a part of a diet aimed at lowering LDL-C and consistently consumed in the daily diet for protective benefits"

". During the LCD, whole eggs were replaced with egg whites, the egg white consumption was reduced to 4.7 egg whites per week, and beef/sausage/or bacon consumption was reduced to 1.1 servings per week. In the LCD, fish consumption went up by 56% per week on average. The number of daily servings of nuts and natural peanut butter (≥ 36 g/serving) increased in the LCD."

(Ferro et al., 2020)

Effects of a Portfolio-Mediterranean Diet and a Mediterranean Diet with or without a Sterol-Enriched Yogurt in Individuals with Hypercholesterolemia

"The Mediterranean Diet (MedDiet) is associated with a reduced low density lipoprotein cholesterol (LDL-C), even in the absence of weight loss (Richard et al., 2011)(Escurriol et al., 2009)"

". LDL-C can be reduced further through a combination of ingredients with cholesterol-lowering properties, such as plant sterols/stanols, viscous plant fibres from oats, barley, and nuts, soymilk, and soy meat analogues. This leads to an LDL-C reduction from 20% to 30% and a triglyceride reduction of 16% in individuals with hyperlipidemia compared with diet alone (Chiavaroli et al., 2018)."

(Chiavaroli et al., 2018)

Portfolio Dietary Pattern and Cardiovascular Disease: A Systematic Review and Meta-analysis of Controlled Trials.

"BACKGROUND

The evidence for the Portfolio dietary pattern, a plant-based dietary pattern that combines recognized cholesterol-lowering foods (nuts, plant protein, viscous fibre, plant sterols), has not been summarized.

OBJECTIVE

To update the European Association for the Study of Diabetes clinical practice guidelines for nutrition therapy, we conducted a systematic review and meta-analysis of controlled trials using GRADE of the effect of the Portfolio dietary pattern on the primary therapeutic lipid target for cardiovascular disease prevention, low-density lipoprotein cholesterol (LDL-C), and other established cardiometabolic risk factors.

METHODS

We searched MEDLINE, EMBASE, and The Cochrane Library through April 19, 2018. We included controlled trials ≥ 8 weeks assessing the effect of the Portfolio dietary pattern on cardiometabolic risk factors compared with an energy-matched control diet free of Portfolio dietary pattern components. Two independent reviewers extracted data and assessed risk of bias. The primary outcome was LDL-C. Data were pooled using the generic inverse-variance method and expressed as mean differences (MDs) with 95% confidence intervals (CIs). Heterogeneity was assessed (Cochran Q statistic) and quantified (I²-statistic). GRADE assessed the certainty of the evidence.

RESULTS

Eligibility criteria were met by 7 trial comparisons in 439 participants with hyperlipidemia, in which the Portfolio dietary pattern was given on a background of a National Cholesterol Education Program (NCEP) Step II diet. The combination of a portfolio dietary pattern and NCEP Step II diet significantly reduced the primary outcome LDL-C by $\sim 17\%$ (MD, -0.73 mmol/L, [95% CI, -0.89 to

-0.56 mmol/L) as well as non-high-density lipoprotein cholesterol, apolipoprotein B, total cholesterol, triglycerides, systolic and diastolic blood pressure, C-reactive protein, and estimated 10-year coronary heart disease (CHD) risk, compared with an NCEP Step 2 diet alone ($p < 0.05$) there was no effect on high-density lipoprotein cholesterol or body weight. The certainty of the evidence was high for LDL-cholesterol and most lipid outcomes and moderate for all other outcomes.

CONCLUSIONS

Current evidence demonstrates that the Portfolio dietary pattern leads to clinically meaningful improvements in LDL-C as well as other established cardiometabolic risk factors and estimated 10-year CHD risk."

(Chang et al., 2025)

[Expert Consensus on the Diagnosis and Management of Carotid Atherosclerotic Plaque: Pathophysiology, Clinical Management, and Preventive Approaches](#)

"For individuals with hypercholesterolemia and those at high cardiovascular risk, daily cholesterol intake should be < 300 mg (approximately the amount in one egg yolk) (Han et al., 2018)(Li et al., 2021)(Kan et al., 2024). If other high-cholesterol foods such as organ meats, red meat, or shrimp are consumed, egg intake should be reduced accordingly. Other recommendations include a daily soybean intake of 25 g, a weekly nut intake of 50-70 g, and a daily dairy intake of 150-300 g. Sugar-sweetened beverages should be avoided. Moderate tea consumption is recommended, with a monthly tea intake of 50-250 g, preferably green tea, while avoiding excessive consumption of strong tea over long periods."

(Kuk, 1987)

[A healthy lifestyle for a healthy heart](#)

"The NIH also recommends that Americans restrict total dietary fat intake from the present level of 40% of calories to 30%. Saturated, monounsaturated, and polyunsaturated fats should each account for no more than 10% of caloric intake, while total cholesterol consumption should be no more than 250-300 mg./day. (For comparison, a single egg yolk contains 252 mg. of cholesterol!) Therefore, disciples of a "healthy heart" diet should skimp on eggs, cream, organ meats (kidney and liver), processed luncheon meats, untrimmed red meat, and foods fried in butter, lard, or fat, and instead eat plenty of fish and poultry, lean meats, fresh fruits and vegetables, sunflower and corn oils, and low-fat dairy products."

Foods and dietary practices to limit or avoid

The main foods to cut back on are saturated fat, trans fat, refined carbs, sugary drinks, and ultra-processed foods. In day-to-day terms, that usually means less butter, cream, fatty and processed meats, fried foods, pastries, chips, and sugary drinks, and more attention to portion size and overall calorie intake. (18 sources)

- **Avoid trans fats as much as possible.** These are among the clearest "avoid" items in cholesterol advice. Common sources include partially hydrogenated oils, some margarines and shortenings, commercial baked goods, and foods made with oils repeatedly heated at high temperature. ([Boyanovsky et al., 2021](#)) ([Rhee et al., 2019](#)) ([Sawhney et al., 2023](#)) ([Cheema, 2012](#))
- **Limit saturated fat.** The main foods to reduce are butter, cream, lard, ghee, palm oil, coconut oil, full-fat dairy, poultry skin, and fatty cuts of meat. Fried foods and foods cooked in butter or animal fat also fit here. ([Diamond et al., 2020](#)) ([Rhee et al., 2019](#)) ([Abbasi et al., 2024](#)) ([Kuk, 1987](#)) ([Cheema, 2012](#))
- **Cut down on red meat, especially processed meat.** Bacon, sausage, processed luncheon meats, and fatty or untrimmed red meat are commonly discouraged because they often bring both saturated fat and excess calories. ([Nestel et al., 2022](#)) ([Boyanovsky et al., 2021](#)) ([Kuk, 1987](#)) ([Abbasi et al., 2024](#))
- **Limit foods high in cholesterol if your LDL is high or your cardiovascular risk is high.** Practical examples include egg yolks, organ meats such as liver and kidney, and some shellfish or shrimp. Several sources use a daily cholesterol limit of less than 300 mg, with less than 200 mg/day used in stricter therapeutic plans. ([Rhee et al., 2019](#)) ([Chang et al., 2025](#)) ([Li et al., 2021](#)) ([Kan et al., 2024](#)) ([Schmier et al., 2009](#)) ([Kuk, 1987](#))
- **Minimize refined starches, added sugars, and sugary drinks.** This includes sweets, pastries, white-flour snack foods, sugar-sweetened beverages, and similar fast-digesting carbs. These foods can worsen triglycerides and make an otherwise "low-fat" diet less helpful. ([Nestel et al., 2022](#)) ([Sawhney et al., 2023](#)) ([Kohli et al., 2014](#)) ([Chang et al., 2025](#)) ([Guo et al., 2025](#)) ([Ma et al., 2024](#))

- **Avoid or sharply limit ultra-processed foods.** This includes many packaged snack foods, industrial breads, chips, and convenience foods that are high in sodium, sugars, trans fats, or additives. ([Garcia et al., 2022](#)) ([Razo-Olvera et al., 2021](#)) ([Boyanovsky et al., 2021](#)) ([Sawhney et al., 2023](#))
- **Limit fried foods.** Fried foods often add saturated fat, trans fat, or excess calories, especially when cooked in reused oil or solid fats. ([Razo-Olvera et al., 2021](#)) ([Cheema, 2012](#)) ([Kuk, 1987](#))
- **Be careful with portion size and overeating.** Even with healthy foods, excess calorie intake can worsen weight, triglycerides, and lipid control. Practical advice includes moderating portions and avoiding routine overeating. ([Li et al., 2021](#)) ([Sawhney et al., 2023](#)) ([Abbasi et al., 2024](#))
- **If triglycerides are also high, avoid alcohol or keep it very low.** This matters more for mixed dyslipidemia or hypertriglyceridemia than for isolated LDL elevation. ([Abbasi et al., 2024](#))
- **In practical "shopping list" terms, foods most people with high cholesterol should usually keep to a minimum are:** butter, cream, full-fat dairy, fatty or processed meats, bacon, organ meats, poultry skin, palm and coconut oils, margarine/shortening with trans fats, fried foods, chips, pastries, industrial breads, sweets, sugary drinks, and other ultra-processed foods. ([Diamond et al., 2020](#)) ([Rhee et al., 2019](#)) ([Nestel et al., 2022](#)) ([Kohli et al., 2014](#))

Evidence

(Boyanovsky et al., 2021)

[Integration of Nutrition Topics in Osteopathic Medical School Curriculum – Shared Experiences](#)

"Although a low-fat diet has a limited effect on LDL and HDL, it may be beneficial in lowering VLDL/triglyceride levels, and patients should be encouraged to consider it, especially the "Mediterranean diet," which uses mostly vegetable and fish-derived fats rich in omega-3 unsaturated fatty acids (Benjamin et al., 2019)"

"foods rich in trans-fats, such as margarine, should be completely excluded. Animal fat (unfortunately bacon) and other foods rich in calories and fats, such as potato and corn chips, should be discouraged."

(Rhee et al., 2019)

[2018 Guidelines for the Management of Dyslipidemia in Korea](#)

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"Eat fiber-rich foods to consume >25 g of dietary fiber"

"Eat diets rich in whole grains and multigrain, beans, vegetables, and fish"

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"Some rich sources of saturated fats include fats of meats, skin of poultry, butter, and palm oil. The major sources of trans fats are hydrogenated oils, such as margarine and shortening, and trans fats are produced during heating oil in high temperature."

(Sawhney et al., 2023)

[CSI clinical practice guidelines for dyslipidemia management: Executive summary](#)

"Increased intake of combination unsaturated fats as in Mediterranean diets is recommended"

"Atherogenic dyslipidemia is best managed by restriction of excess calories, sugars, refined carbohydrates, fried foods, and trans-fats while maintaining ideal body weight. Isolated elevation of LDL cholesterol is managed by restriction of trans-fats and saturated fats"

"The healthiest oils appear to be those high in monounsaturated fats (MUFA) and PUFA with a high N-3/N-6 ratio. The combination of MUFA-and PUFA-containing oils is a good option. Trans fats should be totally avoided"

"Palm oils should be avoided"

"Refined carbohydrates, sugar-sweetened beverages, and processed foods should be strongly restricted and replaced with healthy grains and fiber."

(Cheema, 2012)

Dietary Fats and Heart Disease: From Prenatal to Postnatal Environment

"It is recommended to limit the intake of foods high in saturated fat, trans fat and cholesterol, such as whole-milk dairy products, fatty meats, partially hydrogenated vegetable oils, fried foods and egg yolks. Instead, it is recommended to choose foods low in saturated fat, trans fat and cholesterol. This can be accomplished by enriching the diet in fruits and vegetables, grain products especially from whole grains, lentils and beans and to add either fish or flax powder (ground alsin) to the diet."

(Diamond et al., 2020)

Dietary Recommendations for Familial Hypercholesterolaemia: an Evidence-Free Zone

"DeBeasi, 5 who urged FH individuals to consume 'lean cuts of meat' and to 'remove skin from poultry, select reduced fat cheese and milk, and avoid coconut and palm oils, butter, sour cream, lard, and ghee'. Dietary recommendations in the 2018 AHA cholesterol guidelines stated that FH individuals should follow an LDL-lowering 'heart healthy' diet, which limits animal and vegetable sources of saturated fat, emphasising consumption of 'low-fat dairy EBM analysis: Primary care products and low-fat poultry (without the skin)' ... and nontropical vegetable oils.' 6"

(Abbasi et al., 2024)

New Insights Into the Treatment of Hyperlipidemia: Pharmacological Updates and Emerging Treatments

"Common dietary interventions for hyperlipidemia include decreasing overall food consumption and reducing portion, reducing saturated fat and cholesterol with monounsaturated and polysaturated fats by limiting intake of red meat, choosing lean meat like fish and poultry, and avoiding full-fat dairy products and processed food. Incorporating certain foods with favorable effects on the lipid profile, such as plant sterols and soluble fiber, can lead to a reduction of LDL-C levels by 6%-14%. Additionally, it is advisable to eliminate alcohol in patients with hypertriglyceridemia (Brown et al., 1999)[18]."

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(Nestel et al., 2022)

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(Li et al., 2021)

LncRNA Miat knockdown alleviates endothelial cell injury through regulation of miR-214-3p/Caspase-1 signalling during

atherogenesis

"Atherosclerosis is a common problem in healthy people around the world. Long noncoding RNAs (lncRNAs) play important roles in atherosclerosis. Myocardial infarction-associated transcript (Miat) is a cardiovascular disease-associated lncRNA. Its role and mechanism in atherosclerosis is still not fully clarified. Our study aims to explore the role and mechanism of lncRNA Miat in atherosclerosis. The atherosclerosis models were established both in vitro and in vivo. Real-time PCR was used to measure the expression of lncRNA Miat, miR-214, Caspase-1 and IL-1 β . Western blot was performed to detect the protein expression of Caspase-1. CCK-8 assay, Tunel staining, and flow cytometry analysis were conducted to detect proliferation and apoptosis of human aortic endothelial cells (HAECs), respectively. Oil red O staining and HE staining were used to evaluate the histological changes of the aorta. The results found that lncRNA Miat was upregulated in ox-LDL-induced atherosclerosis model in vitro. The inhibition of lncRNA Miat protects against ox-LDL-induced HAEC injury, presented as increased cell viability and decreased apoptosis. lncRNA Miat and miR-214 has binding site, and CASP1, which encodes Caspase-1, is a target of miR-214. The downregulation of lncRNA Miat increased the expression of miR-214-3p and decreased the expression of Caspase-1, as well as its downstream molecule IL-1 β in HAECs. However, the inhibition of miR-214-3p attenuated the effect of lncRNA Miat downregulation on HAECs. Furthermore, the downregulation of lncRNA Miat alleviated atherosclerosis in ApoE-deficient mice. Correspondingly, the expression of miR-214-3p was upregulated and Caspase-1 was downregulated after knockdown of lncRNA Miat. In conclusion, downregulation of lncRNA Miat exerts a protective effect against atherosclerosis through the regulation miR-214-3p/Caspase-1 signalling pathway. Therefore, the inhibition of lncRNA Miat expression may be an effective strategy in the treatment of atherosclerosis."

(Kan et al., 2024)

Apigenin inhibits proliferation and differentiation of cardiac fibroblasts through AKT/GSK3 β signaling pathway.

"ETHNOPHARMACOLOGICAL RELEVANCE

Salvia miltiorrhiza Bunge (*S. miltiorrhiza*) is an important Traditional Chinese herbal Medicine (TCM) used to treat cardiovascular diseases. Based on the pharmacodynamic substance of *S. miltiorrhiza*, the aim of present study was to investigate the underlying mechanism of *S. miltiorrhiza* against cardiac fibrosis (CF) through a systematic network pharmacology approach, molecular docking and dynamics simulation as well as experimental investigation in vitro.

MATERIALS AND METHODS

A systematic pharmacological analysis was conducted using the Traditional Chinese Medicine Pharmacology (TCMSP) database to screen the effective chemical components of *S. miltiorrhiza*, then the corresponding potential target genes of the compounds were obtained by the Swiss Target Prediction and TCMSP databases. Meanwhile, GeneCards, DisGeNET, OMIM, and TTD disease databases were used to screen CF targets, and a protein-protein interaction (PPI) network of drug-disease targets was constructed on *S. miltiorrhiza*/CF targets by Search Tool for the Retrieval of Interacting Genes/Proteins (STING) database. After that, the component-disease-target network was constructed by software Cytoscape 3.7. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis were performed for the intersection targets between drug and disease. The relationship between active ingredient of *S. miltiorrhiza* and disease targets of CF was assessed via molecular docking and molecular dynamics simulation. Subsequently, the underlying mechanism of the hub compound on CF was experimentally investigated in vitro.

RESULTS

206 corresponding targets to effective chemical components from *S. miltiorrhiza* were determined, and among them, there were 82 targets that overlapped with targets of CF. Further, through PPI analysis, AKT1 and GSK3 β were the hub targets, and which were both enriched in the PI3K/AKT signaling pathway, it was the sub-pathways of the lipid and atherosclerosis pathway. Subsequently, compound-disease-genes-pathways diagram is constructed, apigenin (APi) was a top ingredients and AKT1 (51) and GSK3 β (22) were the hub genes according to the degree value. The results of molecular docking and dynamics simulation showed that APi has strong affinities with AKT and GSK3 β . The results of cell experiments showed that APi inhibited cells viability, proliferation, proteins expression of α -SMA and collagen I/III, phosphorylation of AKT1 and GSK3 β in MCFs induced by TGF β 1.

CONCLUSION

Through a systematic network pharmacology approach, molecular docking and dynamics simulation, and confirmed by in vitro cell experiments, these results indicated that APi interacts with AKT and GSK3 β to disrupt the phosphorylation of AKT and GSK3 β , thereby inhibiting the proliferation and differentiation of MCFs induced by TGF β 1, which providing new insights into the pharmacological mechanism of *S. miltiorrhiza* in the treatment of CF."

(Schmier et al., 2009)

Single food focus dietary guidance: lessons learned from an economic analysis of egg consumption

"Therapeutic lifestyle changes include smoking cessation, increasing physical activity, and modifying one's diet, including keeping daily cholesterol intake less than 200 mg (Iosifescu et al., 2001)."

(Kohli et al., 2014)

Clinician's Guide to the Updated ABCs of Cardiovascular Disease Prevention

"The guideline recommended that patients eat plenty of vegetables, fruits, and whole grains; incorporate low-fat dairy products, poultry, fish, legumes, nontropical vegetable oils, and nuts into their diet; and limit intake of sweets, sugar-sweetened beverages, and red meats. Patients that specifically need to lower their cholesterol levels should reduce saturated and trans fat consumption, with ideally only 5% to 6% of daily calorie intake coming from saturated fat"

". The Mediterranean-style diet (enriched with olive oil, legumes, fish, chicken, nuts, wine, fruits and vegetables and low in artificial sugars, commercial sweets, pastries, butter, margarine, and red meat) also yields heart-healthy benefits. 137"

(Guo et al., 2025)

The influence of psychological intervention-assisted cardiac rehabilitation on kinesophobia in individuals diagnosed with coronary heart disease

"Dietary interventions were customized based on the patient's weight, lipid profile, and glycemic indices, advocating for a diet low in sodium, fat, and sugar, yet rich in fiber, while avoiding high-cholesterol, high-saturated fat, and high-purine foods. Total daily caloric and fluid intake were meticulously regulated, with an emphasis on incorporating fresh fruits and vegetables to sustain optimal weight and nutritional balance (Ma et al., 2024)."

(Ma et al., 2024)

Exploring the effects of an online learning platform in stage III cardiac rehabilitation for individuals with coronary heart disease: Randomized controlled study

"The objective of this study is to assess the influence of blended education methodologies, utilizing an online education platform, among stage III cardiac rehabilitation (CR) patients diagnosed with coronary heart disease (CHD). Between June and August 2021, a cohort of 90 patients diagnosed with CHD, previously discharged from a second-class hospital 1 year earlier, were randomly allocated into 2 groups: the experimental and control groups, with each comprising 45 patients. Patients in the control group received out-of-hospital CR education via WeChat, while those in the experimental group received blended CR education utilizing an online education platform. Following a 24-week period, the self-management behavior and negative emotions of both groups were compared before and after the intervention. The final count of patients in the control and experimental groups was 39 and 37, respectively. Post the intervention, in terms of self-management behavior, the control group achieved an average score of 90.69 ± 7.13 , while the experimental group scored 96.11 ± 5.42 ($P < .05$). Concerning negative emotions, the anxiety scores for the control and experimental groups were 3.03 ± 2.63 and 1.86 ± 1.80 , respectively, and the depression scores were 3.00 (3.00) and 2.00 (3.00), respectively ($P < .05$). The differences in the outcomes mentioned above were statistically significant. The implementation of a blended educational approach utilizing an online platform has resulted in notable improvements in self-management skills and the reduction of negative emotions among patients with CHD. As a result, this educational strategy has demonstrated effectiveness in providing post-discharge CR education for patients with CHD."

(Garcia et al., 2022)

Cardioprotective Diet: A Possibility of Diet Therapy Treatment

"its use recommends higher consumption of fruits, vegetables, wholefiber, such as cereals, and lower consumption of dairy products, fats, and even lean meats"

". This diet also recommends the presence of nutrients (antioxidants and dietary fiber), excluding the ultra-processed foods as recommended in the Food Guide"

". There is also the fourth group, the "red" that is not recommended, because part of this group are the ultra-processed foods, better named ultra-processed foods, which contain large amounts of additives, preservatives and also have high amounts of sodium, sugars and trans fats (Brasil, 2018)."

(Razo-Olvera et al., 2021)

Primary Barriers of Adherence to a Structured Nutritional Intervention in Patients with Dyslipidemia

"Within each nutritional plan, the patients were instructed to avoid the consumption of foods with saturated and trans-fat such as fried foods, margarines, organ meats, and industrialized bread, among others"

". The diets were made up of 40 to 55% carbohydrates, 15 to 20% protein, 30 to 40% fat, and 25 to 35 gr of fiber/day."

(Li et al., 2021)

Influence of Diversity Nursing on Patients' Rehabilitation in Cardiology Treatment

"Cardiology diseases are largely related to the daily eating habits of patients. Excessive intake of high-fat, high-sugar, and high-oil foods can easily cause changes in blood lipids in the body and abnormal blood lipids. Therefore, during the treatment period, we must

pay more attention to and control the patient's diet. According to the patient's condition, personal preferences and past eating habits assist the patient to formulate a scientific diet plan"

"Encourage patients to eat seven to eight percent full at every meal, and not overeating"

"Ask the patient to drink plenty of water, and try to eat digestible foods to avoid constipation."

Practical implementation and special considerations

The most useful way to follow a cholesterol-lowering diet is to make it practical: use regular meal routines, smaller portions, healthy cooking oils, and simple swaps from high-saturated-fat foods to plant foods, fish, and lean proteins. If cholesterol stays high, ask for formal diet counseling, and note that children, people with high triglycerides, and people with kidney or other medical conditions may need a more tailored plan. (13 sources)

In practice, a cholesterol-lowering diet works best when it is turned into a repeatable routine rather than a short-term "diet." Several sources recommend regular eating schedules, portion control, and calorie intake matched to weight goals, since overeating can work against lipid improvement even if food choices are otherwise good. A simple rule is to eat until comfortably satisfied rather than full, keep portions moderate, and use a diet plan that fits the person's usual habits, preferences, weight, lipid pattern, and blood sugar status. ([Kang et al., 2021](#)) ([Li et al., 1, 2021](#)) ([Abbasi et al., 2024](#)) ([Guo et al., 2025](#)) ([Ma et al., 2024](#))

For day-to-day cooking, use food preparation methods that lower saturated fat intake without making the diet hard to follow. Practical examples include cooking with olive, canola, or peanut oil instead of butter or animal fat, choosing lean protein sources, using more cereals, pulses, vegetables, fruit, and milk or dairy in balanced combinations, and avoiding reuse or repeated reheating of oil. Some guidance also suggests rotating vegetable oils to improve overall fatty acid balance. ([Kasiske et al., 2004](#)) ([Sinha et al., 2024](#))

It also helps to think in terms of substitutions. Replace red and processed meat with fish, poultry, beans, lentils, soy foods, or other lean proteins; replace full-fat dairy with low-fat or nonfat versions; and replace foods rich in saturated fat with mono- and polyunsaturated fats. If extra LDL lowering is needed, daily soluble fiber and plant sterols can be added on top of these basic swaps. ([Kim et al., 2021](#)) ([Elisa et al., 2022](#)) ([DiLauro et al., 2023](#)) ([Baratta et al., 2023](#)) ([Abbasi et al., 2024](#))

Lifestyle around the diet matters too. Therapeutic lifestyle advice commonly pairs diet with regular physical activity, enough to help with weight control and overall heart risk, and with fish intake or omega-3 intake as part of the eating pattern. For adults without heart disease, at least 2 servings of fish per week is a common practical target; if triglycerides are high, cutting simple sugars and avoiding alcohol becomes especially important. ([Ito et al., 2009](#)) ([Abbasi et al., 2024](#))

Some people need more structured follow-up. Current guidance described in recent reviews recommends that most patients with hypercholesterolemia start with physician dietary counseling, with referral to a registered dietitian if cholesterol remains high despite initial changes. A reasonable checkpoint is after a few months of consistent diet changes; if LDL is still elevated, stricter saturated fat and cholesterol limits or a more structured pattern may be needed. ([Baratta et al., 2023](#)) ([DiLauro et al., 2023](#))

Special groups may need tailored advice. Children and adolescents with dyslipidemia are often managed with structured CHILD-style diets that set age-appropriate limits on total fat, saturated fat, cholesterol, and trans fat, with lower simple sugars and more fish when triglycerides are high. People with kidney disease may need caution with foods like beans, nuts, seeds, and dairy because potassium or phosphorus can matter even when those foods otherwise fit a heart-healthy pattern. ([Kim, 2025](#)) ([Kasiske et al., 2004](#))

Finally, the plan should be individualized. The broad pattern is clear—less saturated fat, no trans fat, more fiber-rich whole foods, and healthier fats—but the exact version should match the person's medical conditions, culture, food access, and ability to stick with it long term. A sustainable diet that someone can actually follow is more useful than an ideal plan they cannot maintain. ([Li et al., 1, 2021](#)) ([Baratta et al., 2023](#)) ([Guo et al., 2025](#))

Evidence

(Kang et al., 2021)

[Relationship between Dietary Habits and Control of Lipid Profiles in Patients with Dyslipidemia Using Pravastatin](#)

"Therefore, several guidelines have recommended limiting daily cholesterol intake to <300 mg for people with high serum cholesterol levels (Reiner et al., 2011)[6]"

"Patients with dyslipidemia should consume less cholesterol, consume more dairy and protein-rich foods, and follow a regular eating schedule to control lipid profiles."

(Li et al., 2021)

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"Encourage patients to eat seven to eight percent full at every meal, and not overeating"

"Ask the patient to drink plenty of water, and try to eat digestible foods to avoid constipation."

(Abbasi et al., 2024)

[New Insights Into the Treatment of Hyperlipidemia: Pharmacological Updates and Emerging Treatments](#)

"Common dietary interventions for hyperlipidemia include decreasing overall food consumption and reducing portion, reducing saturated fat and cholesterol with monounsaturated and polysaturated fats by limiting intake of red meat, choosing lean meat like fish and poultry, and avoiding full-fat dairy products and processed food. Incorporating certain foods with favorable effects on the lipid profile, such as plant sterols and soluble fiber, can lead to a reduction of LDL-C levels by 6%-14%. Additionally, it is advisable to eliminate alcohol in patients with hypertriglyceridemia (Brown et al., 1999)[18]."

(Guo et al., 2025)

[The influence of psychological intervention-assisted cardiac rehabilitation on kinesiophobia in individuals diagnosed with coronary heart disease](#)

"Dietary interventions were customized based on the patient's weight, lipid profile, and glycemic indices, advocating for a diet low in sodium, fat, and sugar, yet rich in fiber, while avoiding high-cholesterol, high-saturated fat, and high-purine foods. Total daily caloric and fluid intake were meticulously regulated, with an emphasis on incorporating fresh fruits and vegetables to sustain optimal weight and nutritional balance (Ma et al., 2024)."

(Ma et al., 2024)

[Exploring the effects of an online learning platform in stage III cardiac rehabilitation for individuals with coronary heart disease: Randomized controlled study](#)

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(Kasiske et al., 2004)

[Clinical practice guidelines for managing dyslipidemias in kidney transplant patients: a report from the Managing Dyslipidemias in Chronic Kidney Disease Work Group of the National Kidney Foundation Kidney Disease Outcomes Quality Initiative](#)

"Overall, healthy food preparation should be encouraged, such as using peanut, canola, or olive oil in cooking, since these are high in

monounsaturated fats"

"Dietary recommendations may include high-protein foods, with a liberal intake of foods high in saturated fat. However, in the majority of cases, acceptable protein sources low in saturated fat should be encouraged (Table 28). Low-fat dairy products, nuts, seeds, and beans may provide protein, but potassium and phosphorus contents should still be limited."

(Sinha et al., 2024)

[How to Give Dietary Advice to Patients with Kidney Disease?](#)

"Patients with high serum cholesterol should restrict cholesterol to less than 200 mg/day. 20 A judicious amalgamation of cereals, pulses, vegetables, milk, and a variety of vegetable oils aids in maintaining optimal fatty acid composition. Minimization of reusing and reheating of oil is also important. The practice of rotating oils should be encouraged. The recommended combination of oils (groundnut or sesame or rice bran + mustard oil/canola oil/soybean oil) suggested by the National Institute of Nutrition (NIN) may be useful as it helps to maintain a balance to give all the essential nutrients and also maintain the desirable omega -6 to omega 3 - ratios. (Moe et al., 2011)"

(Kim et al., 2021)

[Lifestyle Modification in the Management of Metabolic Syndrome: Statement From Korean Society of CardioMetabolic Syndrome \(KSCMS\)](#)

"Thus, it is recommended to keep saturated fatty acids <10% of total energy and should be further reduced (<7% of energy) in the presence of hypercholesterolemia. (2020) The evidence relevant for the intake of unsaturated fatty acids in dyslipidemia is not yet sufficient, but LDL cholesterol levels decrease when saturated fatty acids are replaced with monounsaturated fatty acids or omega-6 polyunsaturated fatty acids. (Rodríguez, 2014) Trans-fat intake is associated with increased LDL:HDL cholesterol ratio and cardiovascular risk. (Mozaffarian et al., 2006) Hence, it is advisable to avoid transfat intake. A daily fiber intake of 25 g or more is recommended since dietary intake of fibers can reduce the risk of cardiovascular disease by decreasing blood glucose and LDL cholesterol levels. 72)104)"

(Elisa et al., 2022)

[Efficacy of Plant Sterol-Enriched Food for Primary Prevention and Treatment of Hypercholesterolemia: A Systematic Literature Review.](#)

"A healthy diet includes a high intake of fruit, vegetables, nuts, whole grains, lean vegetable or animal protein and fish and reduced intakes of trans fats, red meat, processed meat, refined carbohydrates and sugar-sweetened drinks"

".The National Cholesterol Education Program Adult Treatment Panel III recommends combined dietary therapy with a low rate of both saturated and total fat, a total cholesterol concentration lower than 200 mg/dL and consumption of soluble fiber and plant sterols/phytosterols (PS), a dose of 2 g/day of PSs being more effective."

(DiLauro et al., 2023)

[The Healthy Eating Assessment Tool \(HEAT\): A Simplified 10-Point Assessment of CHILd-2 Dietary Compliance for Children and Adolescents with Dyslipidemia](#)

"A low-fat, low-sugar, and high-fiber diet that relies primarily on vegetables and fruits, whole grains, lean meats, fish, lentils and legumes, and low-and non-fat dairy products is recommended"

".saturated fat is to be limited to 8-10% of daily kcal/EER, while mono and polyunsaturated fat consumption is encouraged up to 20% of daily kcal/EER. Trans fat is to be avoided as much as possible, and dietary cholesterol is to be limited to <300 mg/day. The remaining 70% of dietary energy should be composed of 15-20% proteins and 50-55% carbohydrates, with a focus on fiber-rich foods such as vegetables, fruit, and whole grains. If dyslipidemia persists after 3 months, the CHILd-2 is recommended, which suggests a further restriction of saturated fat to <7% of daily kcal/EER, monounsaturated fat to 10% of daily kcal/EER, and limitation of dietary cholesterol to <200 mg/day (Table 1)."

(Baratta et al., 2023)

[Challenges in Improving Adherence to Diet and Drug Treatment in Hypercholesterolemia Patients](#)

"A good adherence to a Mediterranean-type low-fat diet and healthful lifestyle is associated with lower serum cholesterol levels and reduced odds of having hypercholesterolemia. Therefore, current guidelines recommend initial dietary counseling by physicians for most patients with hypercholesterolemia and referral to a registered dietitian for those with persistent hypercholesterolemia (2020)6,19]. The main dietary recommendation is to minimize the consumption of saturated fats to lower LDL-C, as well as other therapeutic dietary options, such as increasing plant stanols/sterols and increasing viscous fiber (2020)."

(Ito et al., 2009)

[Maintaining Cardiovascular Health in Patients With Mixed Dyslipidemia Optimizing the Management of Hypertriglyceridemia and Non-HDL Cholesterol](#)

"Recommended therapeutic lifestyle changes include increased physical activity (moderate exercise sufficient to expend ≥ 200 kcal per day); increased intake of omega-3 fatty acids, plant stanols/sterols, and soluble fiber; a caloric intake limited to that needed to maintain an appropriate body weight and prevent weight gain; and limited dietary intake of saturated fats, cholesterol, and trans fatty acids. Limiting saturated fat intake to $< 7\%$ of total calories and dietary cholesterol to < 200 mg per day should be the primary recommendation in all patients with elevated LDL-C levels. Other therapeutic options include intake of 2 gm per day of plant sterols/stanols and 10-25 gm per day of soluble fiber. 32,(Ellis et al., 2004)"

"The American Heart Association (AHA) recommends that all individuals increase the amount of fish incorporated into the diet as a source of omega-3 fatty acids that may lower CHD risk. People without coronary disease should be consuming at least 2 servings of fish per week. In patients with preexisting CHD, the AHA recommends the equivalent of 1 gm per day of omega-3 fatty acids-a level shown to decrease the risk of cardiovascular death."

(Kim, 2025)

Dyslipidemia in Children and Adolescents: Current Insights and Updated Treatment Approaches

"For children and adolescents with dyslipidemia, transitioning to a low-fat diet is recommended to reduce LDL-C levels"

".CHILD 1 limits total fat intake to 25-30% of total calories, saturated fat to $<10\%$, and cholesterol to <300 mg per day"

".CHILD 2-LDL is intended for patients with hyperlipidemia characterized by elevated LDL-C, which limits total fat intake to 25-30% of total calories, saturated fat to $<7\%$, cholesterol to <200 mg/day, and trans-fat to $<1\%$. CHILD 2-TG is indicated for patients with hypertriglyceridemia. This approach focuses on reducing simple carbohydrates, while increasing the amount of complex carbohydrates. CHILD 2-TG also recommends consuming dietary fish to increase omega-3 fatty acid intake."